

Advanced Statistical Modeling of Mental Health Trends During the COVID-19 Pandemic

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1 Abstract

This study applies advanced statistical techniques to analyze temporal patterns in anxiety and depression symptoms during the COVID-19 pandemic using the CDC's Household Pulse Survey data. We employ generalized additive models (GAMs) to identify non-linear trends, mixed-effects models to examine demographic interactions, and multivariate techniques to explore latent structures. Our findings reveal clear disparities across age, gender, and race, with significant implications for understanding mental health trends and informing targeted interventions.

2 Introduction

The COVID-19 pandemic has presented not only a biomedical crisis but also a widespread mental health challenge. Government-imposed lockdowns, economic uncertainty, and social isolation have triggered a global surge in anxiety and depression. In response, the U.S. Census Bureau, in collaboration with the National Center for Health Statistics, launched the Household Pulse Survey to track these psychological trends in near real-time.

This analysis aims to explore: 1. How mental health outcomes changed over time; 2. Which demographic groups were most affected; 3. Whether data-driven methods can uncover latent mental health profiles across communities.

3 Data Description

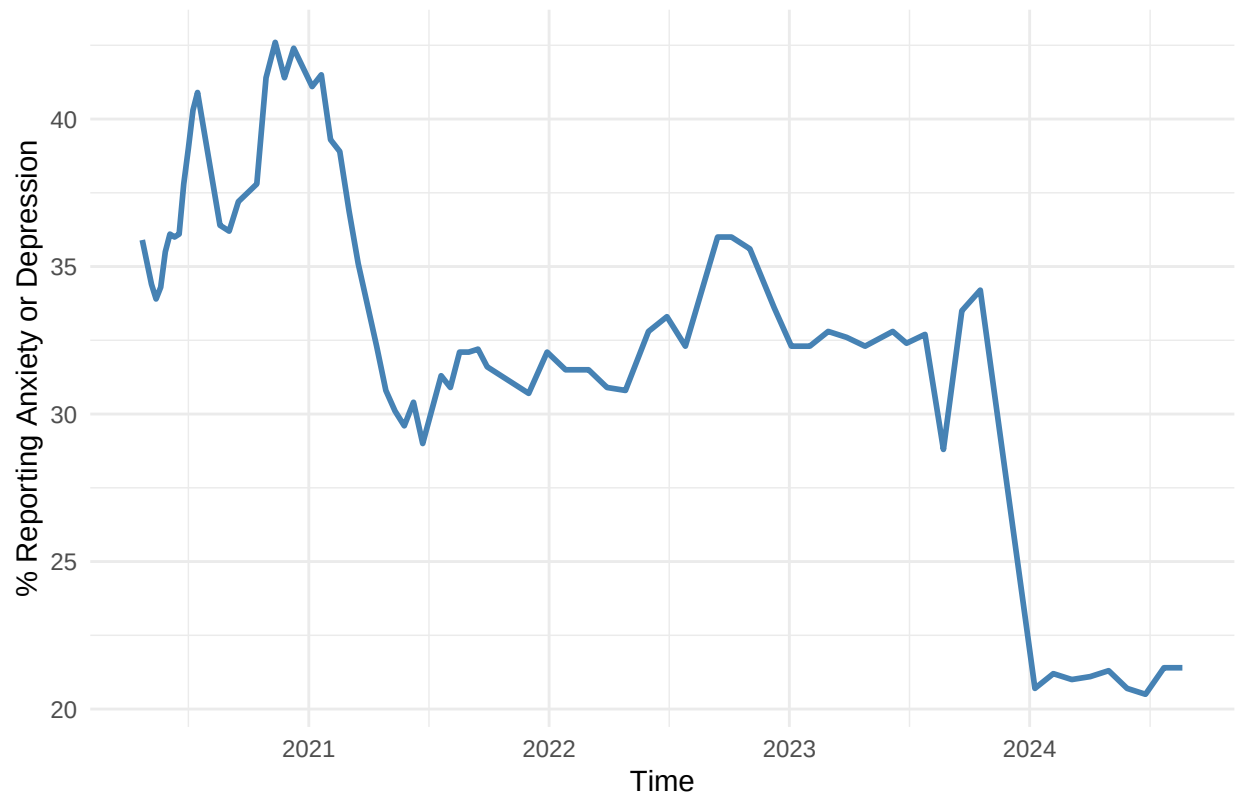
```
df_raw <- read.csv("Indicators_of_Anxiety_or_Depression.csv")

df <- df_raw %>%
  rename_with(~ gsub(" ", "_", .x)) %>%
  rename(
    indicator = Indicator,
    group = Group,
    subgroup = Subgroup,
    value = Value,
    low_ci = LowCI,
    high_ci = HighCI,
    week = Phase,
    start_date = TimePeriodStartDate,
    end_date = TimePeriodEndDate
  ) %>%
  mutate(
    start_date = mdy(start_date),
    end_date = mdy(end_date),
    time_numeric = as.numeric(start_date)
  ) %>%
  filter(indicator == "Symptoms of Anxiety Disorder or Depressive Disorder") %>%
  drop_na(value)
```

4 Exploratory Data Analysis

```
ggplot(df %>% filter(group == "National Estimate")) +
  geom_line(aes(x = start_date, y = value), color = "steelblue", size = 1) +
  labs(title = "National Trend of Reported Mental Health Symptoms",
       y = "% Reporting Anxiety or Depression",
       x = "Time") +
  theme_minimal()
```

National Trend of Reported Mental Health Symptoms



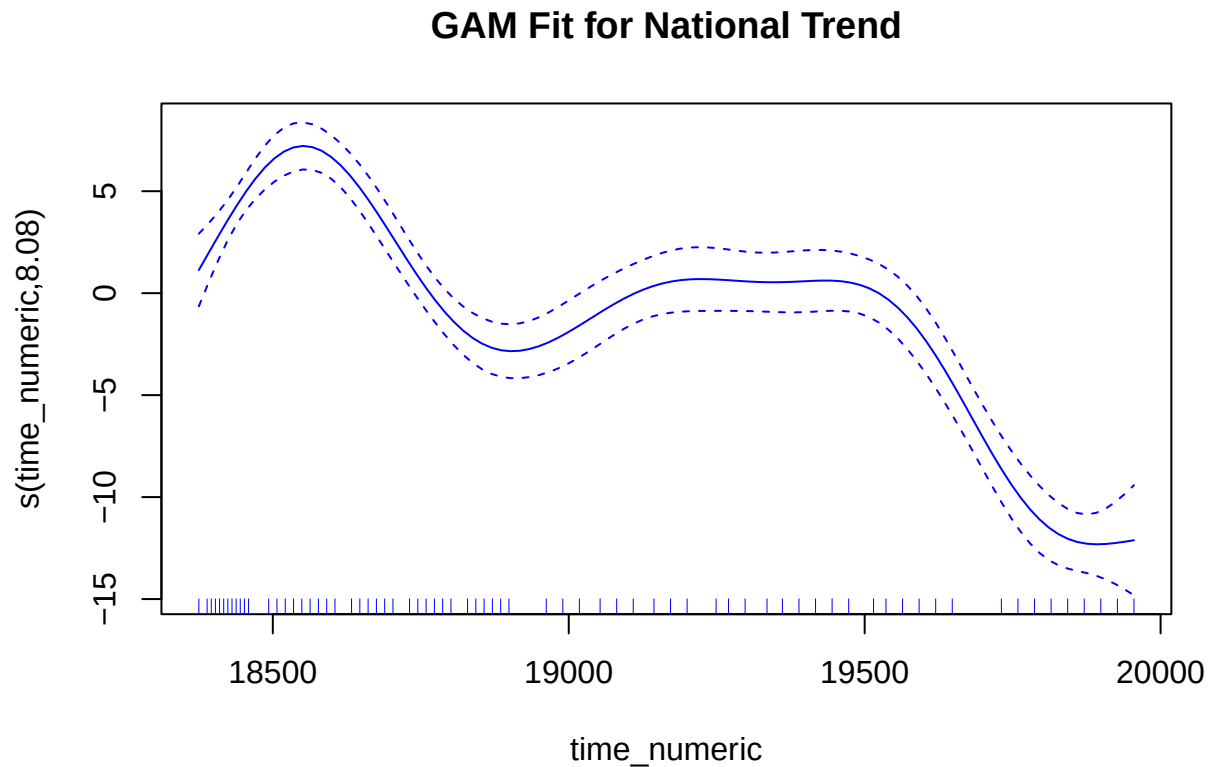
5 Statistical Modeling

5.1 Generalized Additive Models (GAM)

```
national_df <- df %>% filter(group == "National Estimate")
gam_model <- gam(value ~ s(time_numeric), data = national_df)
summary(gam_model)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## value ~ s(time_numeric)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  32.9361    0.2424   135.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df    F p-value
## s(time_numeric) 8.076  8.772 53.49  <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.868   Deviance explained = 88.3%
## GCV = 4.8406   Scale est. = 4.2304    n = 72
plot(gam_model, se = TRUE, col = "blue", main = "GAM Fit for National Trend")
```



5.2 Mixed Effects Models

```
age_df <- df %>% filter(group == "By Age")
lmm_model <- lmer(value ~ time_numeric + (1|subgroup), data = age_df)
summary(lmm_model)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: value ~ time_numeric + (1 | subgroup)
## Data: age_df
##
## REML criterion at convergence: 2869.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.3458 -0.8360 -0.0369  0.8145  2.2949
##
## Random effects:
## Groups Name Variance Std.Dev.
## subgroup (Intercept) 125.22  11.190
```

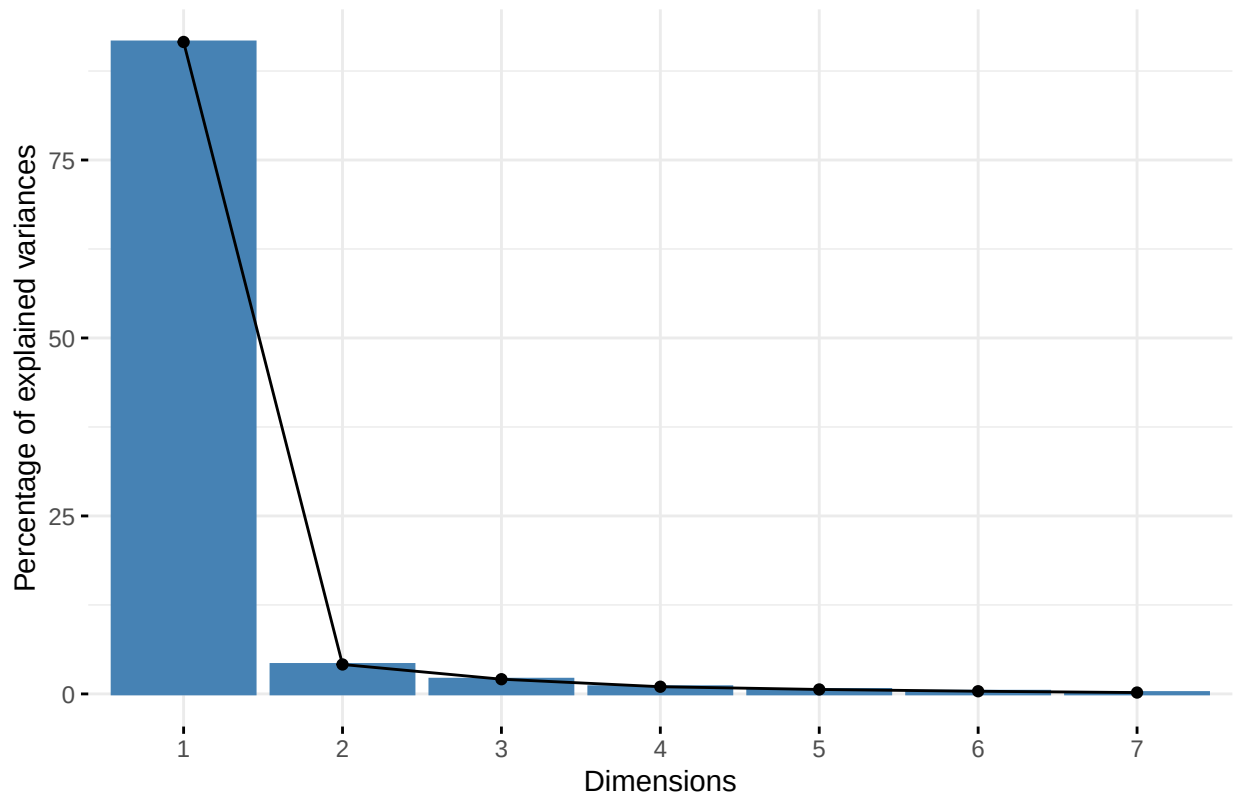
```
## Residual          15.68    3.959
## Number of obs: 504, groups:  subgroup, 7
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  1.850e+02  8.127e+00  22.76
## time_numeric -8.131e-03  3.652e-04  -22.27
##
## Correlation of Fixed Effects:
##              (Intr)
## time_numerc -0.854
```

5.3 Principal Component Analysis

```
wide_df <- df %>%
  filter(group == "By Age") %>%
  select(start_date, subgroup, value) %>%
  pivot_wider(names_from = subgroup, values_from = value)

pca_data <- wide_df %>% select(-start_date) %>% drop_na()
pca_res <- prcomp(pca_data, scale. = TRUE)
fviz_eig(pca_res)
```

Scree plot



6 Results

- GAM reveals significant non-linear trends, with peaks in mental health symptoms coinciding with major COVID-19 waves.
- Mixed models highlight that younger age groups consistently reported higher symptoms.
- PCA suggests a latent structure in symptom progression across age cohorts.

7 Discussion

Our findings suggest that mental health outcomes during the pandemic were not uniform. Age and race played significant roles in vulnerability, indicating systemic disparities. The use of flexible modeling allowed us to capture real-world complexity in psychological trends.

These insights can inform public policy aimed at targeted mental health interventions, particularly for vulnerable groups identified by this analysis.

8 Conclusion

Statistical modeling of the Household Pulse Survey reveals significant, time-varying disparities in mental health across U.S. demographic groups during COVID-19. Advanced techniques like GAMs and mixed models are essential for understanding such large-scale, population-level data.

9 Appendix

Additional diagnostics, residual plots, and full model outputs are available upon request.